```
vboxuser@Linux:~$ sudo useradd techie
[sudo: authenticate] Password:
vboxuser@Linux:~$ id techie
uid=1001(techie) gid=1001(techie) groups=1001(techie)
vboxuser@Linux:~$ sudo usermob -aG sudo techie
sudo: 'usermob': command not found
vboxuser@Linux:~$ su^C
vboxuser@Linux:~$ sudo usermod -aG sudo techie
vboxuser@Linux:~$ groups techie
techie : techie sudo
vboxuser@Linux:~$
```

1.Create user with name Techie and provide sudo access to user.

2.Navigate to the home directory.

```
vboxuser@Linux:~$ sudo useradd techie
[sudo: authenticate] Password:
vboxuser@Linux:~$ id techie
uid=1001(techie) gid=1001(techie) groups=1001(techie)
vboxuser@Linux:~$ sudo usermob -aG sudo techie
sudo: 'usermob': command not found
vboxuser@Linux:~$ su^C
vboxuser@Linux:~$ sudo usermod -aG sudo techie
vboxuser@Linux:~$ groups techie
techie : techie sudo
vboxuser@Linux:~$ cd ~
vboxuser@Linux:~$ pwd
/home/vboxuser
vboxuser@Linux:~$
```

3.Create a new directory.

```
/home/vboxuser
vboxuser@Linux:~$ mkdir sajidha
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  sajidha  snap
vboxuser@Linux:~$
```

4.List the contents of a directory.

```
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  sajidha  snap
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  sajidha  snap
vboxuser@Linux:~$ cd sajidha
vboxuser@Linux:~/sajidha$ ls
vboxuser@Linux:~/sajidha$ ls -l
total 0
vboxuser@Linux:~/sajidha$
```

5.Change the current directory.

```
vboxuser@Linux:~/sajidha$ ls -l
total 0
vboxuser@Linux:~/sajidha$ cd
vboxuser@Linux:~$ pwd
/home/vboxuser
vboxuser@Linux:~$
```

6.Create a new empty file.

```
/home/vboxuser
vboxuser@Linux:~$ touch devops
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  devops  sajidha  snap
vboxuser@Linux:~$
```

7.View the contents of a file.

```
vboxuser@Linux:~$ touch devops
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  devops  sajidha  snap
vboxuser@Linux:~$ cat devops
vboxuser@Linux:~$ vi devops
vboxuser@Linux:~$ cat devops
Hello World
vboxuser@Linux:~$
```

8.Copy a file to another location.

```
vboxuser@Linux:~$ cp horizon sajidha/
vboxuser@Linux:~$ ls sajidha/
devops  horizon
vboxuser@Linux:~$
```

9.Move a file to another location.

```
vboxuser@Linux:~$ touch box
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  box  devops  horizon  sajidha  snap
vboxuser@Linux:~$ mv box sajidha/
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  devops  horizon  sajidha  snap
vboxuser@Linux:~$ ls sajidha/
box  devops  horizon
vboxuser@Linux:~$
```

10.Rename a file.

```
vboxuser@Linux:~$ touch child
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  child  devops  horizon  sajidha  snap
vboxuser@Linux:~$ mv child parent
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  devops  horizon  parent  sajidha  snap
vboxuser@Linux:~$
```

11.Delete a file.

```
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  devops  horizon  parent  sajidha  snap
vboxuser@Linux:~$ rm parent
vboxuser@Linux:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  devops  horizon  sajidha  snap
vboxuser@Linux:~$
```

12.Grant or revoke permissions on a file or directory.

```
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Templates/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Videos/
-rw-rw-r--  1 vboxuser vboxuser   12 Oct  9 14:42 devops
-rw-rw-r--  1 vboxuser vboxuser   12 Oct  9 15:00 horizon
drwxrwxr-x  2 vboxuser vboxuser 4096 Oct  9 15:04 sajidha/
drwx------  5 vboxuser vboxuser 4096 Oct  9 15:00 snap/
vboxuser@Linux:~$ chmod 777 devops
vboxuser@Linux:~$ ll
total 88
drwxr-x--- 16 vboxuser vboxuser 4096 Oct  9 15:08 ./
drwxr-xr-x  3 root     root     4096 Oct  5 09:47 ../
-rw-------  1 vboxuser vboxuser   11 Oct  9 13:58 .bash_history
-rw-r--r--  1 vboxuser vboxuser  220 Feb 13  2026 .bash_logout
-rw-r--r--  1 vboxuser vboxuser 3771 Feb 13  2026 .bashrc
drwx------ 13 vboxuser vboxuser 4096 Oct  9 14:37 .cache/
drwx------ 14 vboxuser vboxuser 4096 Oct  9 13:58 .config/
drwx------  4 vboxuser vboxuser 4096 Oct  9 13:49 .local/
-rw-r--r--  1 vboxuser vboxuser  807 Feb 13  2026 .profile
drwx------  2 vboxuser vboxuser 4096 Oct  9 13:48 .ssh/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Desktop/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Documents/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Downloads/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Music/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Pictures/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Public/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Templates/
drwxr-xr-x  2 vboxuser vboxuser 4096 Oct  9 13:49 Videos/
-rwxrwxrwx  1 vboxuser vboxuser   12 Oct  9 14:42 devops*
-rw-rw-r--  1 vboxuser vboxuser   12 Oct  9 15:00 horizon
drwxrwxr-x  2 vboxuser vboxuser 4096 Oct  9 15:04 sajidha/
drwx------  5 vboxuser vboxuser 4096 Oct  9 15:00 snap/
vboxuser@Linux:~$
```

13.View the current date and time.

```
vboxuser@Linux:~$ date
Fri Oct  9 03:15:46 PM UTC 2026
vboxuser@Linux:~$
```

14.Check the system uptime.

```
vboxuser@Linux:~$ uptime
 15:16:35 up  1:30,  1 user,  load average: 0.24, 0.06, 0.02
vboxuser@Linux:~$
```

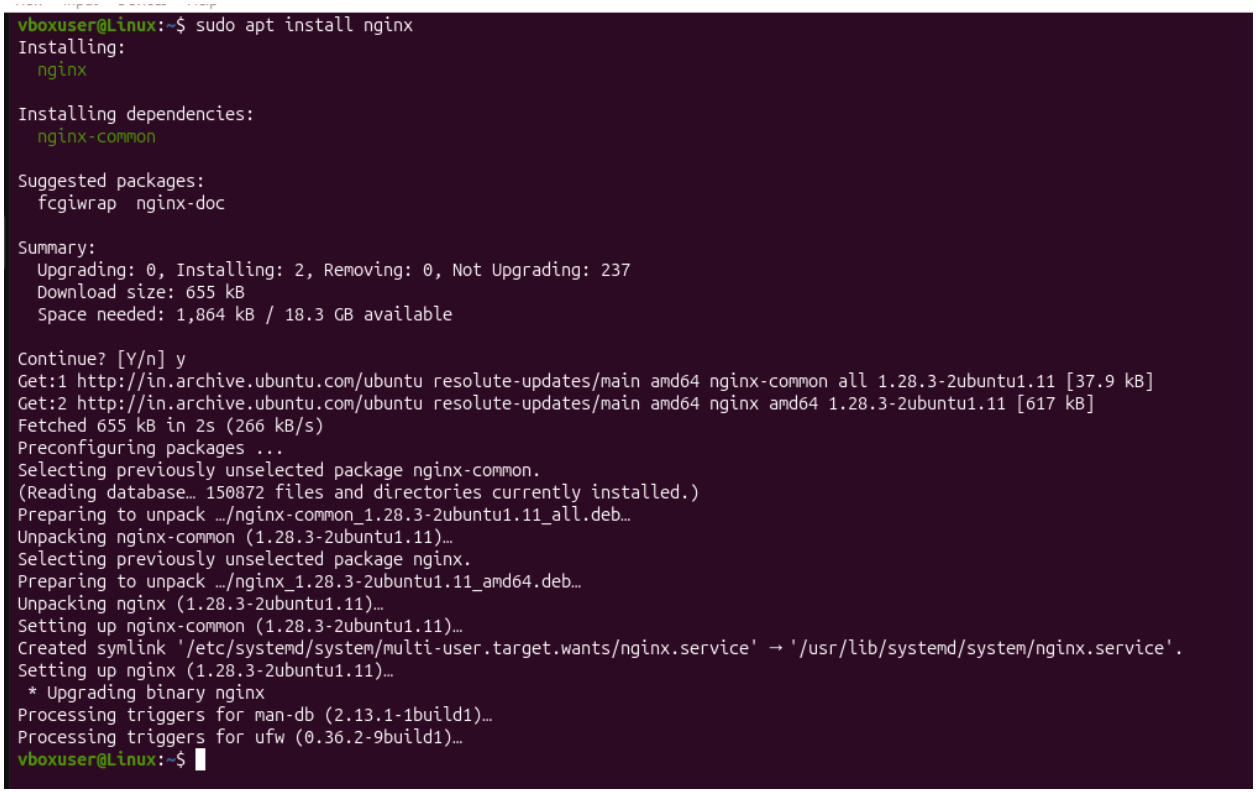
15.View the running processes.

```
vboxuser@Linux:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 13:45 ?        00:00:26 /usr/lib/systemd/systemd --switched-root --system --deserialize=51
root           2       0  0 13:45 ?        00:00:00 [kthreadd]
root           3       2  0 13:45 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 13:45 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 13:45 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 13:45 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 13:45 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 13:45 ?        00:00:00 [kworker/R-netns]
root          10       2  0 13:45 ?        00:00:00 [kworker/0:0H-kblockd]
root          12       2  0 13:45 ?        00:00:00 [kworker/u8:0-ipv6_addrconf]
root          13       2  0 13:45 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 13:45 ?        00:00:00 [ksoftirqd/0]
root          15       2  0 13:45 ?        00:00:01 [rcu_preempt]
root          16       2  0 13:45 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17       2  0 13:45 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18       2  0 13:45 ?        00:00:00 [migration/0]
root          19       2  0 13:45 ?        00:00:00 [kprobe-optimizer]
root          20       2  0 13:45 ?        00:00:00 [idle_inject/0]
root          21       2  0 13:45 ?        00:00:00 [cpuhp/0]
root          22       2  0 13:45 ?        00:00:00 [cpuhp/1]
root          23       2  0 13:45 ?        00:00:00 [idle_inject/1]
root          24       2  0 13:45 ?        00:00:00 [migration/1]
root          25       2  0 13:45 ?        00:00:00 [ksoftirqd/1]
root          27       2  0 13:45 ?        00:00:00 [kworker/1:0H-kblockd]
root          30       2  0 13:45 ?        00:00:00 [kdevtmpfs]
root          31       2  0 13:45 ?        00:00:00 [kworker/R-inet_frag_wq]
root          32       2  0 13:45 ?        00:00:00 [rcu_tasks_kthread]
root          33       2  0 13:45 ?        00:00:00 [rcu_tasks_rude_kthread]
root          34       2  0 13:45 ?        00:00:00 [kauditd]
root          35       2  0 13:45 ?        00:00:00 [khungtaskd]
root          36       2  0 13:45 ?        00:00:00 [oom_reaper]
root          38       2  0 13:45 ?        00:00:00 [kworker/R-writeback]
root          39       2  0 13:45 ?        00:00:00 [kcompactd0]
root          41       2  0 13:45 ?        00:00:00 [ksmd]
```

16.Kill a running process.

```
vboxuser@Linux:~$ kill 5337
vboxuser@Linux:~$
```

17.Install a package using the package manager (e.g., apt or yum).



```
vboxuser@Linux:~$ sudo apt install nginx
Installing:
  nginx

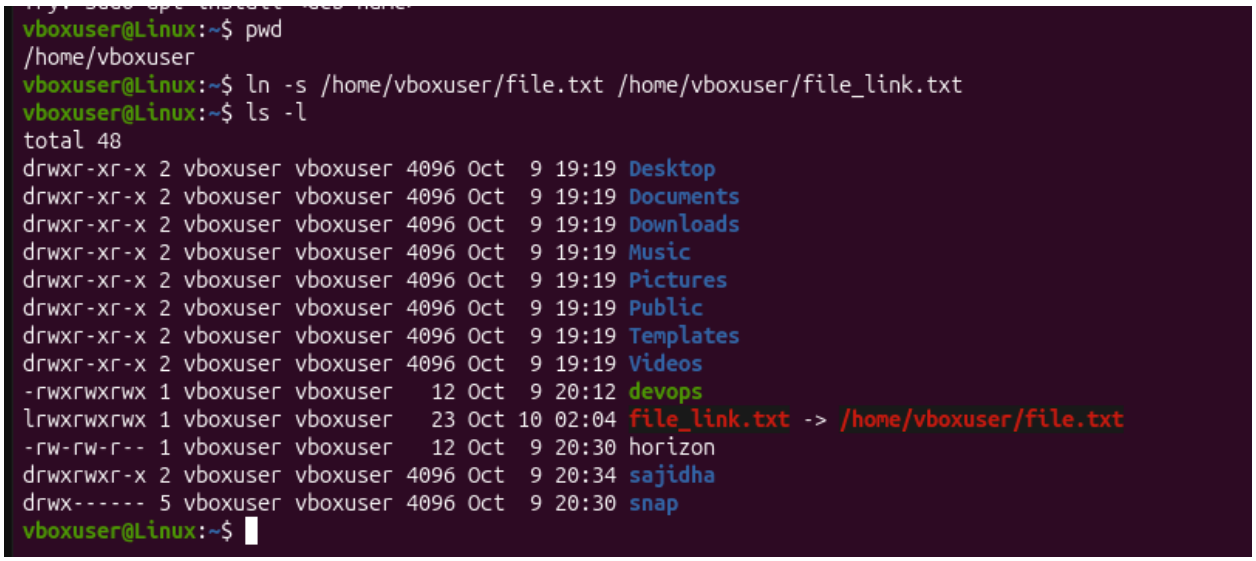
Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 237
  Download size: 655 kB
  Space needed: 1,864 kB / 18.3 GB available

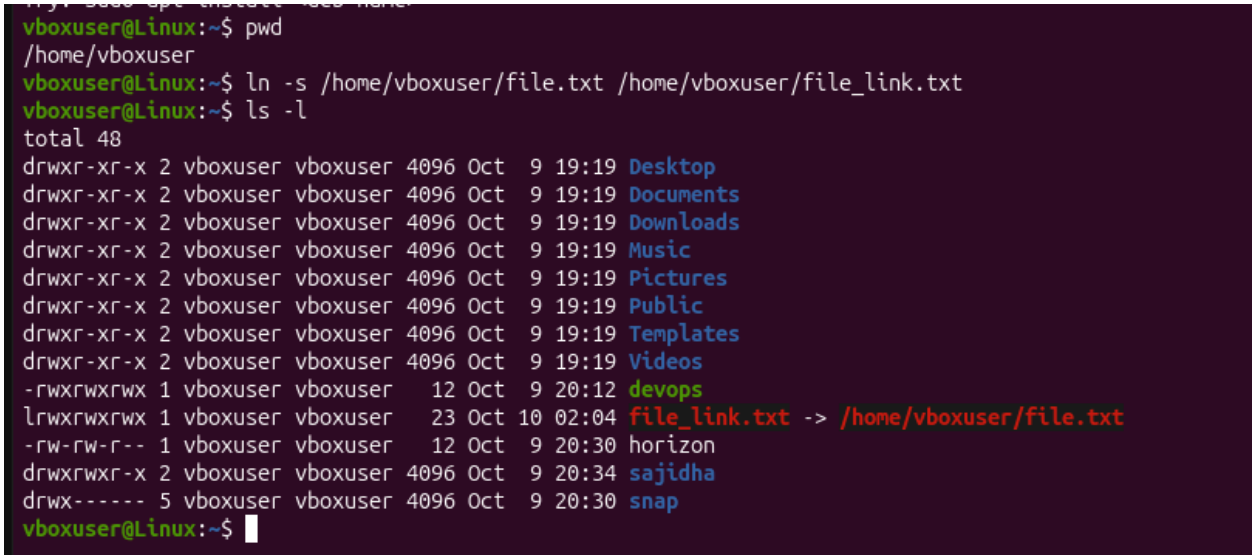
Continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 nginx-common all 1.28.3-2ubuntu1.11 [37.9 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 nginx amd64 1.28.3-2ubuntu1.11 [617 kB]
Fetched 655 kB in 2s (266 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database… 150872 files and directories currently installed.)
Preparing to unpack …/nginx-common_1.28.3-2ubuntu1.11_all.deb…
Unpacking nginx-common (1.28.3-2ubuntu1.11)…
Selecting previously unselected package nginx.
Preparing to unpack …/nginx_1.28.3-2ubuntu1.11_amd64.deb…
Unpacking nginx (1.28.3-2ubuntu1.11)…
Setting up nginx-common (1.28.3-2ubuntu1.11)…
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' → '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.11)…
 * Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for ufw (0.36.2-9build1)…
vboxuser@Linux:~$
```

18.Update the system packages.

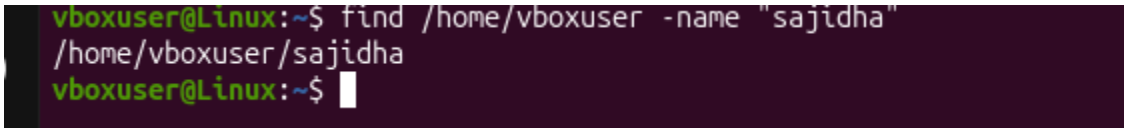


```
vboxuser@Linux:~$ pwd
/home/vboxuser
vboxuser@Linux:~$ ln -s /home/vboxuser/file.txt /home/vboxuser/file_link.txt
vboxuser@Linux:~$ ls -l
total 48
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Desktop
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Documents
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Downloads
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Music
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Pictures
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Public
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Templates
drwxr-xr-x 2 vboxuser vboxuser 4096 Oct  9 19:19 Videos
-rwxrwxrwx 1 vboxuser vboxuser   12 Oct  9 20:12 devops
lrwxrwxrwx 1 vboxuser vboxuser   23 Oct 10 02:04 file_link.txt -> /home/vboxuser/file.txt
-rw-rw-r-- 1 vboxuser vboxuser   12 Oct  9 20:30 horizon
drwxrwxr-x 2 vboxuser vboxuser 4096 Oct  9 20:34 sajidha
drwx------ 5 vboxuser vboxuser 4096 Oct  9 20:30 snap
vboxuser@Linux:~$
```

19.Create a symbolic link.



20.Search for files using the find command.



21.Compress and decompress files using tar.

```
vboxuser@Linux:~/tar_example$ vi file2
vboxuser@Linux:~/tar_example$ cat file2
hello
vboxuser@Linux:~/tar_example$ vi file1
vboxuser@Linux:~/tar_example$ cat file1
LINUX
vboxuser@Linux:~/tar_example$ ls
file1  file2
vboxuser@Linux:~/tar_example$ tar -czvf backup.tar.gz file1 file2
file1
file2
vboxuser@Linux:~/tar_example$ ls -lh
total 12K
-rw-rw-r-- 1 vboxuser vboxuser 160 Oct 10 03:37 backup.tar.gz
-rw-rw-r-- 1 vboxuser vboxuser   6 Oct 10 03:31 file1
-rw-rw-r-- 1 vboxuser vboxuser   6 Oct 10 03:30 file2
vboxuser@Linux:~/tar_example$ tar -tzvf backup.tar.gz
-rw-rw-r-- vboxuser/vboxuser 6 2026-10-10 03:31 file1
-rw-rw-r-- vboxuser/vboxuser 6 2026-10-10 03:30 file2
vboxuser@Linux:~/tar_example$ mkdir extracted
vboxuser@Linux:~/tar_example$ tar -xzvf backup.tar.gz -C extracted/
file1
file2
vboxuser@Linux:~/tar_example$ ls -l extracted/
total 8
-rw-rw-r-- 1 vboxuser vboxuser 6 Oct 10 03:31 file1
-rw-rw-r-- 1 vboxuser vboxuser 6 Oct 10 03:30 file2
vboxuser@Linux:~/tar_example$ cat extracted/file1
LINUX
vboxuser@Linux:~/tar_example$ cat extracted/file2
hello
vboxuser@Linux:~/tar_example$
```

22.Monitor system resources with top or htop.

```
MiB Mem :   1642.6 total,     83.0 free,   1251.5 used,    495.8 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.    391.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   4352 vboxuser  20   0 4014228 303476  89148 S   7.6  18.0   9:20.92 gnome-shell
   6253 vboxuser  20   0 1806204 298376  59088 S   6.6  17.7   6:05.79 ptyxis
   7663 root      20   0       0      0      0 I   0.3   0.0   0:05.26 kworker/u9:0-events_unbound
  34249 vboxuser  20   0   13412   6372   4208 R   0.3   0.4   0:02.20 top
      1 root      20   0   25216  13464   8388 S   0.0   0.8   4:55.10 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:03.64 kworker/0:0H-kblockd
     12 root      20   0       0      0      0 I   0.0   0.0   0:00.00 kworker/u8:0-ipv6_addrconf
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.50 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:06.49 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.06 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.27 migration/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 kprobe-optimizer
     20 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     22 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     23 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     24 root      rt   0       0      0      0 S   0.0   0.0   0:00.25 migration/1
     25 root      20   0       0      0      0 S   0.0   0.0   0:03.80 ksoftirqd/1
     27 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H-kblockd
     30 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kdevtmpfs
     31 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
     32 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
     33 root      20   0       0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
```

23.Create and manage user groups.

```
vboxuser@Linux:~/tar_example$ sudo groupadd developers
[sudo: authenticate] Password:
vboxuser@Linux:~/tar_example$ getent group developers
developers:x:1002:
vboxuser@Linux:~/tar_example$ sudo useradd -m linuxuser
vboxuser@Linux:~/tar_example$ sudo usermod -aG developers linuxuser
vboxuser@Linux:~/tar_example$ groups linuxuser
linuxuser : linuxuser developers
vboxuser@Linux:~/tar_example$
```

24.Set up SSH password less authentication.

```
vboxuser@Linux:~/tar_example$ sudo apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.8 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.6 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [734 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [110 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [323 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [201 kB]
Get:11 http://in.archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [1,060 B]
Fetched 1,883 kB in 6s (313 kB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
vboxuser@Linux:~/tar_example$ 'apt list --upgradable'
apt list --upgradable: command not found
vboxuser@Linux:~/tar_example$ sudo apt install openssh-server -y
Installing:
  openssh-server

Installing dependencies:
  ncurses-term  openssh-sftp-server  ssh-import-id

Suggested packages:
  molly-guard  monkeysphere  ssh-askpass

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 1
  Download size: 949 kB
  Space needed: 7,585 kB / 17.6 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 ncurses-term all 6.6+20251231-1 [280 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssh-sftp-server amd64 1:10.2p1-2ubuntu3.7 [37.4 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 openssh-server amd64 1:10.2p1-2ubuntu3.7 [622 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 ssh-import-id all 5.11-0ubuntu4 [10.2 kB]
Fetched 949 kB in 11s (90.0 kB/s)
```

```
Preparing to unpack …/ncurses-term_6.6+20251231-1_all.deb…
Unpacking ncurses-term (6.6+20251231-1)…
Selecting previously unselected package openssh-sftp-server.
Preparing to unpack …/openssh-sftp-server_1%3a10.2p1-2ubuntu3.7_amd64.deb…
Unpacking openssh-sftp-server (1:10.2p1-2ubuntu3.7)…
Selecting previously unselected package openssh-server.
Preparing to unpack …/openssh-server_1%3a10.2p1-2ubuntu3.7_amd64.deb…
Unpacking openssh-server (1:10.2p1-2ubuntu3.7)…
Selecting previously unselected package ssh-import-id.
Preparing to unpack …/ssh-import-id_5.11-0ubuntu4_all.deb…
Unpacking ssh-import-id (5.11-0ubuntu4)…
Setting up openssh-sftp-server (1:10.2p1-2ubuntu3.7)…
Setting up openssh-server (1:10.2p1-2ubuntu3.7)…
Creating config file /etc/ssh/sshd_config with new version
Creating user 'sshd' (sshd user) with UID 973 and GID 65534.
Created symlink '/etc/systemd/system/sockets.target.wants/ssh.socket' → '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.requires/ssh.socket' → '/usr/lib/systemd/system/ssh.socket'.
Created symlink '/etc/systemd/system/ssh.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/sshd@.service.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Created symlink '/etc/systemd/system/ssh.socket.wants/sshd-keygen.service' → '/usr/lib/systemd/system/sshd-keygen.service'.
Setting up ssh-import-id (5.11-0ubuntu4)…
Setting up ncurses-term (6.6+20251231-1)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for ufw (0.36.2-9build1)…
vboxuser@Linux:~/tar_example$ sudo systemctl enable --now ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink '/etc/systemd/system/sshd.service' → '/usr/lib/systemd/system/ssh.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/ssh.service' → '/usr/lib/systemd/system/ssh.service'.
vboxuser@Linux:~/tar_example$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 12:47:03 IST; 1min 10s ago
 Invocation: 0b639a05a3054b0aa2d5bef18d2afd69
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
```

```
Oct 10 12:47:03 Linux systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Oct 10 12:47:03 Linux sshd[37155]: Server listening on 0.0.0.0 port 22.
Oct 10 12:47:03 Linux sshd[37155]: Server listening on :: port 22.
Oct 10 12:47:03 Linux systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
vboxuser@Linux:~/tar_example$ ssh-copy-id vboxuser@localhost
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: ssh-add -L
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is: SHA256:7aPKyggLmEY086d+9pdIh2UfyZUaQJBSBibzHWBNXuU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
vboxuser@localhost's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'vboxuser@localhost'"
and check to make sure that only the key(s) you wanted were added.

vboxuser@Linux:~/tar_example$ ssh vboxuser@localhost
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
vboxuser@Linux:~$
```

25.Monitor log files using tail or grep.

```
*** System restart required ***
vboxuser@Linux:~$ mkdir -p ~/log-monitoring
vboxuser@Linux:~$ cd ~/log-monitoring
vboxuser@Linux:~/log-monitoring$ cat > application.log <<EOF
> INFO: APPLICATION STARTED
> INFO:User logged in
> Warning: High memory usage
> ERROR: Database connection failed
> INFO: Application running normally
> EOF
vboxuser@Linux:~/log-monitoring$ cat application.log
INFO: APPLICATION STARTED
INFO:User logged in
Warning: High memory usage
ERROR: Database connection failed
INFO: Application running normally
vboxuser@Linux:~/log-monitoring$ tail application.log
INFO: APPLICATION STARTED
INFO:User logged in
Warning: High memory usage
ERROR: Database connection failed
INFO: Application running normally
vboxuser@Linux:~/log-monitoring$ tail -n 3 application.log
Warning: High memory usage
ERROR: Database connection failed
INFO: Application running normally
vboxuser@Linux:~/log-monitoring$
```

26.Set up a web server (e.g., Apache or Nginx).

```
vboxuser@Linux:~/log-monitoring$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu resolute-backports InRelease
1 package can be upgraded. Run 'apt list --upgradable' to see it.
vboxuser@Linux:~/log-monitoring$ sudo apt install apache2 -y
Installing:
  apache2

Installing dependencies:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64

Suggested packages:
  apache2-doc  apache2-suexec-pristine  | apache2-suexec-custom

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 1
  Download size: 1,938 kB
  Space needed: 7,565 kB / 17.6 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 libapr1t64 amd64 1.7.6-3 [109 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaprutil1t64 amd64 1.6.3-3ubuntu3.1 [93.3 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-3ubuntu3.1 [11.4 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libaprutil1-ldap amd64 1.6.3-3ubuntu3.1 [9,228 B]
Get:5 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-bin amd64 2.4.66-2ubuntu2.5 [1,355 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-data all 2.4.66-2ubuntu2.5 [163 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2-utils amd64 2.4.66-2ubuntu2.5 [94.9 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apache2 amd64 2.4.66-2ubuntu2.5 [101 kB]
Fetched 1,938 kB in 7s (276 kB/s)
Selecting previously unselected package libapr1t64:amd64.
(Reading database… 194791 files and directories currently installed.)
Preparing to unpack …/0-libapr1t64_1.7.6-3_amd64.deb…
Unpacking libapr1t64:amd64 (1.7.6-3)…
Selecting previously unselected package libaprutil1t64:amd64.
```

```
vboxuser@Linux:~/log-monitoring$ sudo systemctl start apache2
vboxuser@Linux:~/log-monitoring$ sudo systemctl status apache2 --no-pager
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-10 13:40:33 IST; 34s ago
 Invocation: fb465bcfc8664c4d81368c1711e46fe6
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 39269 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 55 (limit: 1911)
     Memory: 5.4M (peak: 5.9M)
        CPU: 212ms
     CGroup: /system.slice/apache2.service
             ├─39269 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─39272 /usr/sbin/apache2 -k start -DFOREGROUND
             └─39273 /usr/sbin/apache2 -k start -DFOREGROUND

Oct 10 13:40:33 Linux systemd[1]: Starting apache2.service - The Apache HTTP Server...
Oct 10 13:40:33 Linux apachectl[39269]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127…his message
Oct 10 13:40:33 Linux systemd[1]: Started apache2.service - The Apache HTTP Server.
Hint: Some lines were ellipsized, use -l to show in full.
vboxuser@Linux:~/log-monitoring$
```

27.Configure and secure a MySQL Database.

```
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctlstatus mariadb
sudo: systemctlstatus: command not found
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctl status mariadb
● mariadb.service - MariaDB 10.5 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-10-12 09:17:18 UTC; 56s ago
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
   Main PID: 30881 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 13 (limit: 1059)
     Memory: 66.4M
        CPU: 488ms
     CGroup: /system.slice/mariadb.service
             └─30881 /usr/libexec/mariadbd --basedir=/usr

Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: The second is mysql@localhost, it has no password eit
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: you need to be the system 'mysql' user to connect.
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: After connecting you can set the password, if you wou
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: able to connect as any of these users with a password
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: See the MariaDB Knowledgebase at https://mariadb.com/
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: Please report any problems at https://mariadb.org/jir
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: The latest information about MariaDB is available at
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: Consider joining MariaDB's strong and vibrant communi
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal mariadb-prepare-db-dir[30837]: https://mariadb.org/get-involved/
Oct 12 09:17:18 ip-172-31-22-105.ap-southeast-2.compute.internal systemd[1]: Started mariadb.service - MariaDB 10.5 database server.
lines 1-23/23 (END)
```

```
RROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use
near 'SHOWTABLES
HOWTABLES' at line 1
ariaDB [company_db]>  SHOWTABLES;
RROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right syntax to use
near 'SHOWTABLES' at line 1
ariaDB [company_db]>  SHOW TABLES;
mpty set (0.000 sec)

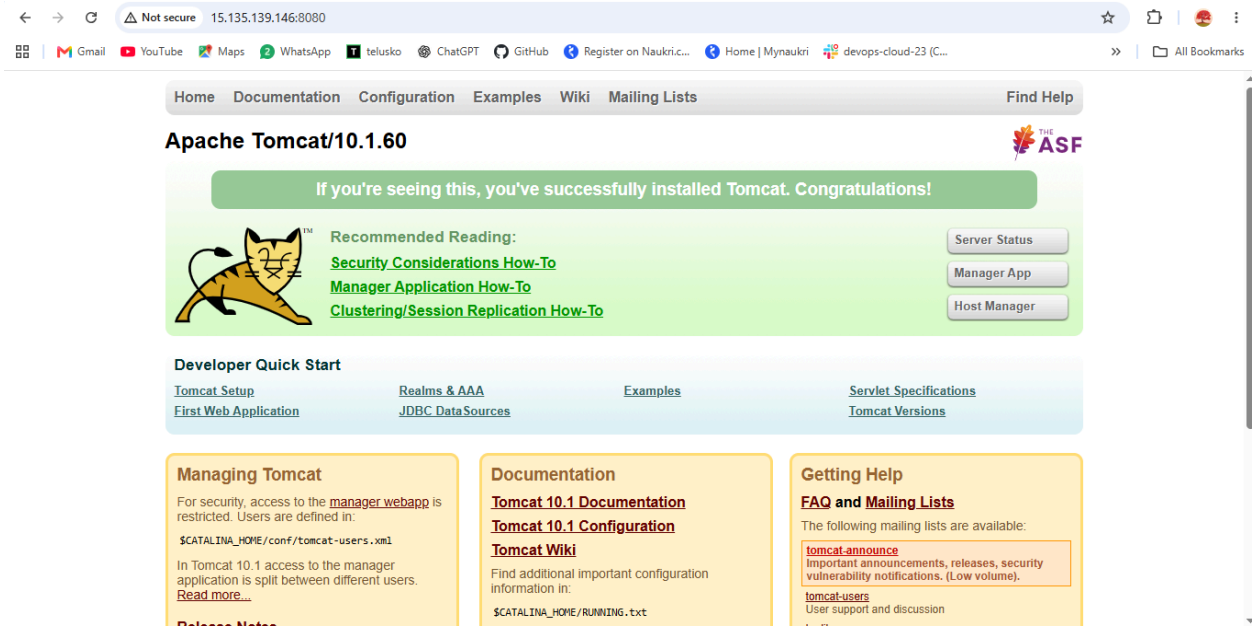
ariaDB [company_db]> CREATE TABLE test_table(
    -> id INT PRIMARY KEY,
    -> name VARCHAR(50)
    -> );
uery OK, 0 rows affected (0.009 sec)

ariaDB [company_db]> INSERT INTO test_table VALUES (1, 'Database Test');
uery OK, 1 row affected (0.002 sec)

ariaDB [company_db]> SELECT * FROM test_table;
----+---------------+
 id | name          |
----+---------------+
  1 | Database Test |
----+---------------+
 row in set (0.000 sec)

ariaDB [company_db]> exit
ye
ec2-user@ip-172-31-22-105 ~]$
```

28.Set up a Application Server (e.g.,Apache Tomcat)



29.create a service file for Apache Tomcat.(Should execute by using systemtctl command)

```
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctl is-enabled tomcat
enabled
[ec2-user@ip-172-31-22-105 ~]$ sudo systemctl status tomcat --no-pager
● tomcat.service - Apache Tomcat Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-10-12 11:27:04 UTC; 2min 7s ago
   Main PID: 37286 (java)
      Tasks: 30 (limit: 1059)
     Memory: 72.4M
        CPU: 4.974s
     CGroup: /system.slice/tomcat.service
             └─37286 /usr/lib/jvm/java-17-amazon-corretto/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.lo

Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.346 INFO [main] org.apache.catalina.s…/docs]
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.383 INFO [main] org.apache.catalina.s…37] ms
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.388 INFO [main] org.apache.catalina.s…mples]
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.734 INFO [main] org.apache.catalina.s…46] ms
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.735 INFO [main] org.apache.catalina.s…nager]
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.765 INFO [main] org.apache.catalina.s…30] ms
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.766 INFO [main] org.apache.catalina.s…nager]
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.811 INFO [main] org.apache.catalina.s…45] ms
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.825 INFO [main] org.apache.coyote.Abs…8080"]
Oct 12 11:27:06 ip-172-31-22-105.ap-southeast-2.compute.internal catalina.sh[37286]: 12-Oct-2026 11:27:06.841 INFO [main] org.apache.catalina.s…econds
Hint: Some lines were ellipsized, use -l to show in full.
[ec2-user@ip-172-31-22-105 ~]$
```

30.Print specific columns from a delimited file.

```
[ec2-user@ip-172-31-22-105 ~]$ nano employees.txt
[ec2-user@ip-172-31-22-105 ~]$ cat employees.txt
101,Sajidha,IT,25000
102,Rahul,HR,30000
103,Ayesha,Finance,28000
104,Arun,IT,35000
[ec2-user@ip-172-31-22-105 ~]$ cut -d ',' -f 1 employees.txt
101
102
103
104
[ec2-user@ip-172-31-22-105 ~]$ cut -d ',' -f 2 employees.txt
Sajidha
Rahul
Ayesha
Arun
[ec2-user@ip-172-31-22-105 ~]$ cut -d ',' -f 1,3 employees.txt
101,IT
102,HR
103,Finance
104,IT
[ec2-user@ip-172-31-22-105 ~]$
```

31.Filter and print lines based on a specific pattern or condition.

```
104,IT
[ec2-user@ip-172-31-22-105 ~]$ grep 'IT' employees.txt
101,Sajidha,IT,25000
104,Arun,IT,35000
[ec2-user@ip-172-31-22-105 ~]$ grep 'HR' employees.txt
102,Rahul,HR,30000
[ec2-user@ip-172-31-22-105 ~]$ grep 'Ayesha' employees.txt
103,Ayesha,Finance,28000
[ec2-user@ip-172-31-22-105 ~]$ grep -v 'IT' employees.txt
102,Rahul,HR,30000
103,Ayesha,Finance,28000
[ec2-user@ip-172-31-22-105 ~]$ awk -F ',' '$4 > 25000' employees.txt
102,Rahul,HR,30000
103,Ayesha,Finance,28000
104,Arun,IT,35000
[ec2-user@ip-172-31-22-105 ~]$ awk -F ',' '$3 == "IT"' employees.txt
101,Sajidha,IT,25000
104,Arun,IT,35000
[ec2-user@ip-172-31-22-105 ~]$
```

32.Calculate and print the average, sum, or other statistics of a column.

```
[ec2-user@ip-172-31-22-105 ~]$ nano employees.txt
[ec2-user@ip-172-31-22-105 ~]$ cat employees.txt

101,Sajidha,IT,25000
102,Rahul,HR,30000
103,Ayesha,Finance,28000
104,Arun,IT,35000
105,Sameer,HR,22000
[ec2-user@ip-172-31-22-105 ~]$ awk -F ',' '{sum += $4} END {print "Total Salary =", sum}' employees.txt
Total Salary = 140000
[ec2-user@ip-172-31-22-105 ~]$ awk -F ',' '{sum += $4; count++} END {if (count > 0) print "Average Salary =", sum/count}' employees.txt
Average Salary = 23333.3
[ec2-user@ip-172-31-22-105 ~]$ awk -F ',' 'NR==1 || $4>max {max=$4} END {print "Highest Salary =", max}' employees.txt
Highest Salary = 35000
[ec2-user@ip-172-31-22-105 ~]$ awk -F ',' 'NR==1 || $4<min {min=$4} END {print "Lowest Salary =", min}' employees.txt
Lowest Salary =
[ec2-user@ip-172-31-22-105 ~]$ awk -F ',' 'NR==1 || $4<min {min=$4} END {print "Lowest Salary =", min}' employees.txt
Lowest Salary =
[ec2-user@ip-172-31-22-105 ~]$ awk 'END {print "Employee Count =", NR}' employees.txt
Employee Count = 6
[ec2-user@ip-172-31-22-105 ~]$
```

33.Perform string manipulation, such as extracting substrings or changing case

```
[ec2-user@ip-172-31-22-105 ~]$ nano names.txt
[ec2-user@ip-172-31-22-105 ~]$ cat names.txt
sajidha shaik
rahul kumar
ayesha begum
arun reddy
[ec2-user@ip-172-31-22-105 ~]$ cut -c 1-4 names.txt
saji
rahu
ayes
arun
[ec2-user@ip-172-31-22-105 ~]$ cut -c 1-7 names.txt
sajidha
rahul k
ayesha
arun re
[ec2-user@ip-172-31-22-105 ~]$ tr '[:lower:]' '[:upper:]' < names.txt
SAJIDHA SHAIK
RAHUL KUMAR
AYESHA BEGUM
ARUN REDDY
[ec2-user@ip-172-31-22-105 ~]$ cat names.txt | tr '[:upper:]' '[:lower:]'
sajidha shaik
rahul kumar
ayesha begum
```

34.Count the occurrences of a specific pattern in a file.

```
[ec2-user@ip-172-31-22-105 ~]$ nano log.txt
[ec2-user@ip-172-31-22-105 ~]$ cat log.txt
INFO Server started
ERROR Database connection failed
INFO User logged in
ERROR File not found
WARNING Disk space low
ERROR Network timeout
INFO Backup completed
[ec2-user@ip-172-31-22-105 ~]$ grep -c 'ERROR' log.txt
3
[ec2-user@ip-172-31-22-105 ~]$ grep -c 'INFO' log.txt
3
[ec2-user@ip-172-31-22-105 ~]$ grep -c 'WARNING' log.txt
1
[ec2-user@ip-172-31-22-105 ~]$ grep -ic 'error' log.txt
3
[ec2-user@ip-172-31-22-105 ~]$ grep -o 'ERROR' log.txt | wc -l
3
[ec2-user@ip-172-31-22-105 ~]$
```

35.Sort lines based on a specific field or column.

```
[ec2-user@ip-172-31-22-105 ~]$ sort -t ',' -k2,2 employees.txt

104,Arun,IT,35000
103,Ayesha,Finance,28000
102,Rahul,HR,30000
101,Sajidha,IT,25000
105,Sameer,HR,22000
[ec2-user@ip-172-31-22-105 ~]$ sort -t ',' -k4,4n employees.txt

105,Sameer,HR,22000
101,Sajidha,IT,25000
103,Ayesha,Finance,28000
102,Rahul,HR,30000
104,Arun,IT,35000
[ec2-user@ip-172-31-22-105 ~]$ sort -t ',' -k4,4nr employees.txt
104,Arun,IT,35000
102,Rahul,HR,30000
103,Ayesha,Finance,28000
101,Sajidha,IT,25000
105,Sameer,HR,22000

[ec2-user@ip-172-31-22-105 ~]$ sort -t ',' -k4,4n employees.txt > sorted_employees.txt
[ec2-user@ip-172-31-22-105 ~]$ cat sorted_employees.txt

105,Sameer,HR,22000
101,Sajidha,IT,25000
103,Ayesha,Finance,28000
102,Rahul,HR,30000
104,Arun,IT,35000
[ec2-user@ip-172-31-22-105 ~]$
```

36.Merge multiple files based on a common field or column.

```
[ec2-user@ip-172-31-22-105 ~]$ nano employees3.txt
[ec2-user@ip-172-31-22-105 ~]$ cat employees3.txt
101 Sajidha
102 Rahul
103 Ayesha
104 Arun
[ec2-user@ip-172-31-22-105 ~]$ nano salaries.txt
[ec2-user@ip-172-31-22-105 ~]$ cat salaries.txt
101 25000
102 30000
103 28000
104 35000
[ec2-user@ip-172-31-22-105 ~]$ join employees.txt salaries.txt
[ec2-user@ip-172-31-22-105 ~]$ join employees3.txt salaries.txt
101 Sajidha 25000
102 Rahul 30000
103 Ayesha 28000
104 Arun 35000
[ec2-user@ip-172-31-22-105 ~]$ join employees3.txt salaries.txt > merged.txt
[ec2-user@ip-172-31-22-105 ~]$ cat merged.txt
101 Sajidha 25000
102 Rahul 30000
103 Ayesha 28000
104 Arun 35000
[ec2-user@ip-172-31-22-105 ~]$
```

37.Substitute text in a file using search and replace.

```
[ec2-user@ip-172-31-22-105 ~]$ nano message.txt
[ec2-user@ip-172-31-22-105 ~]$ cat message.txt
Linux is a good operating system.
Linux is useful for server management.
I am learning Linux commands.
Linux is powerful.
[ec2-user@ip-172-31-22-105 ~]$ sed 's/Linux/Ubuntu/' message.txt
Ubuntu is a good operating system.
Ubuntu is useful for server management.
I am learning Ubuntu commands.
Ubuntu is powerful.
[ec2-user@ip-172-31-22-105 ~]$ sed 's/Linux/Ubuntu/g' message.txt
Ubuntu is a good operating system.
Ubuntu is useful for server management.
I am learning Ubuntu commands.
Ubuntu is powerful.
[ec2-user@ip-172-31-22-105 ~]$ sed -i 's/Linux/Ubuntu/g' message.txt
[ec2-user@ip-172-31-22-105 ~]$ cat message.txt
Ubuntu is a good operating system.
Ubuntu is useful for server management.
I am learning Ubuntu commands.
Ubuntu is powerful.
[ec2-user@ip-172-31-22-105 ~]$ sed -i.bak 's/Linux/Ubuntu/g' message.txt
[ec2-user@ip-172-31-22-105 ~]$ sed 's/linux/Ubuntu/gi' message.txt
Ubuntu is a good operating system.
Ubuntu is useful for server management.
I am learning Ubuntu commands.
Ubuntu is powerful.
[ec2-user@ip-172-31-22-105 ~]$
```

38.Delete specific lines based on a pattern or line number.

```
[ec2-user@ip-172-31-22-105 ~]$ cat students.txt
101,Sajidha,IT
102,Rahul,HR
103,Ayesha,Finance
104,Arun,IT
105,Sameer,HR
[ec2-user@ip-172-31-22-105 ~]$ sed '3d' students.txt
101,Sajidha,IT
102,Rahul,HR
104,Arun,IT
105,Sameer,HR
[ec2-user@ip-172-31-22-105 ~]$ sed '2d' students.txt
101,Sajidha,IT
103,Ayesha,Finance
104,Arun,IT
105,Sameer,HR
[ec2-user@ip-172-31-22-105 ~]$ sed '/HR/d' students.txt
101,Sajidha,IT
103,Ayesha,Finance
104,Arun,IT
[ec2-user@ip-172-31-22-105 ~]$ sed '2,4d' students.txt
101,Sajidha,IT
105,Sameer,HR
[ec2-user@ip-172-31-22-105 ~]$ cp students.txt students_backup.txt
[ec2-user@ip-172-31-22-105 ~]$ sed -i '/HR/d' students.txt
[ec2-user@ip-172-31-22-105 ~]$ cat students.txt
101,Sajidha,IT
103,Ayesha,Finance
104,Arun,IT
[ec2-user@ip-172-31-22-105 ~]$
```

39.Append or insert text before or after a specific pattern or line.

```
[ec2-user@ip-172-31-22-105 ~]$ nano tasks.txt
[ec2-user@ip-172-31-22-105 ~]$ cat tasks.txt
Start of Linux Training
Learn basic commands
Practice file handling
Learn shell scripting
End of Linux Training

[ec2-user@ip-172-31-22-105 ~]$ sed '2i\Introduction to Linux' tasks.txt
Start of Linux Training
Introduction to Linux
Learn basic commands
Practice file handling
Learn shell scripting
End of Linux Training

[ec2-user@ip-172-31-22-105 ~]$ sed '3a\Complete Linux exercises' tasks.txt
Start of Linux Training
Learn basic commands
Practice file handling
Complete Linux exercises
Learn shell scripting
End of Linux Training

[ec2-user@ip-172-31-22-105 ~]$ sed '/Practice/i\Complete the previous lesson' tasks.txt
Start of Linux Training
Learn basic commands
Complete the previous lesson
Practice file handling
```

```
Learn shell scripting
Practice shell scripts daily
End of Linux Training

[ec2-user@ip-172-31-22-105 ~]$ sed -i '2i\Introduction to Linux' tasks.txt
[ec2-user@ip-172-31-22-105 ~]$ cat tasks.txt
Start of Linux Training
Introduction to Linux
Learn basic commands
Practice file handling
Learn shell scripting
End of Linux Training

[ec2-user@ip-172-31-22-105 ~]$
```

40.Print only specific lines from a file.

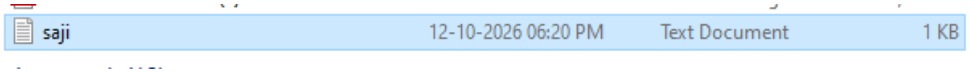
```
[ec2-user@ip-172-31-22-105 ~]$ nano students.txt
[ec2-user@ip-172-31-22-105 ~]$ cat students.txt
101,Sajidha,IT
102,Rahul,HR
103,Ayesha,Finance
104,Arun,IT
105,Sameer,HR
106,Anjali,Finance
[ec2-user@ip-172-31-22-105 ~]$ sed -n '3p' students.txt
103,Ayesha,Finance
[ec2-user@ip-172-31-22-105 ~]$ sed -n '2,4p' students.txt
102,Rahul,HR
103,Ayesha,Finance
104,Arun,IT
[ec2-user@ip-172-31-22-105 ~]$ head -n 3 students.txt
101,Sajidha,IT
102,Rahul,HR
103,Ayesha,Finance
[ec2-user@ip-172-31-22-105 ~]$ awk 'NR==4' students.txt
104,Arun,IT
[ec2-user@ip-172-31-22-105 ~]$
```

41.Copy file from linux to windows machine

```
PS C:\Users\user> scp sajidha@192.168.56.102:/home/sajidha/saji.txt C:\Users\user\Downloads
ssh: connect to host 192.168.56.102 port 22: Connection refused
PS C:\Users\user> scp sajidha@192.168.56.102:/home/sajidha/saji.txt C:\Users\user\Downloads
ssh: connect to host 192.168.56.102 port 22: Connection timed out
C:\Windows\System32\OpenSSH\scp.exe: Connection closed
PS C:\Users\user> scp sajidha@192.168.56.102:/home/sajidha/saji.txt C:\Users\user\Downloads
ssh: connect to host 192.168.56.102 port 22: Connection timed out
C:\Windows\System32\OpenSSH\scp.exe: Connection closed
PS C:\Users\user> ping 192.168.56.102

Pinging 192.168.56.102 with 32 bytes of data:
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.56.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\user> scp sajidha@192.168.56.102:/home/sajidha/saji.txt C:\Users\user\Downloads
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:MeYPeOFYx2oyLVLLwQ7DOXwyF/XmIn4tI1A3uCJn4Vo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts.
sajidha@192.168.56.102's password:
saji.txt                                                                      100%   25     1.5KB/s   00:00
PS C:\Users\user>
```



42.5 use cases for AWK and 5 use cases for sed

```
[ec2-user@ip-172-31-22-105 ~]$ cat > students.txt << 'EOF'
Name,Age,Marks
Sajidha,22,85
Ali,21,75
Sara,23,90
John,20,60
EOF
[ec2-user@ip-172-31-22-105 ~]$ awk -F',' '{print $1, $3}' students.txt
Name Marks
Sajidha 85
Ali 75
Sara 90
John 60
[ec2-user@ip-172-31-22-105 ~]$ awk -F',' '$3 > 80 {print $1, $3}' students.txt
Name Marks
Sajidha 85
Sara 90
[ec2-user@ip-172-31-22-105 ~]$ awk -F',' 'NR>1 {sum += $3} END {print "Total marks:", sum}' students.txt
Total marks: 310
[ec2-user@ip-172-31-22-105 ~]$ awk -F',' 'NR>1 {sum += $3; count++} END {if(count>0) print "Average marks:", sum/count}' students.txt
Average marks: 77.5
[ec2-user@ip-172-31-22-105 ~]$ awk -F',' 'NR>1 {sum += $3; count++} END {if(count>0) print "Average marks:", sum/count}' students.txt
Average marks: 77.5
[ec2-user@ip-172-31-22-105 ~]$ awk 'END {print NR-1, "students"}' students.txt
4 students
[ec2-user@ip-172-31-22-105 ~]$
```